Ocean quahog (*Arctica islandica*)

Summary Scores

- **Sensitivity:** Very High
- **Exposure:** Moderate
- **Vulnerability:** High

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	4	3	
Complexity in Reproductive Strategy	2.44	2.02	
Dispersal of Early Life Stages	1.8	2.5	
Egg and Larval Survival and Settlement Requirements	2.72	2.16	
Historical Variability in Temperature	2.48	2.6	
Juvenile Habitat Requirements	1.32	2.8	
Other Stressors	2.72	2.65	
Population Growth Rate	3.96	2.86	
Prey Specificity	1.24	2.9	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.6	2.15	
Spawning Cycle	2.04	2.6	
Stock Size/Status	1.88	2.4	
Total Sensitivity | Certainty	**Very High (4) | 0.71**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	3.02	0.96	
Bottom Oxygen	3.13	0.87	
Bottom Salinity	2.96	0.78	
Bottom Aragonite Solubility	1.05	0.32	
Net primary production	1.01	0.05	
Downward Flux of Particulate Organic Carbon	1.16	0.25	
Total Exposure	**Moderate (2.46)**	**0.90**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**High (9.85)**	**3.58**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Ocean quahog. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Very High	Very High	None
Exposure	High	Moderate	Decrease
Vulnerability	Very High	High	Decrease
Directional Effect	Negative	Negative	None

Stocks

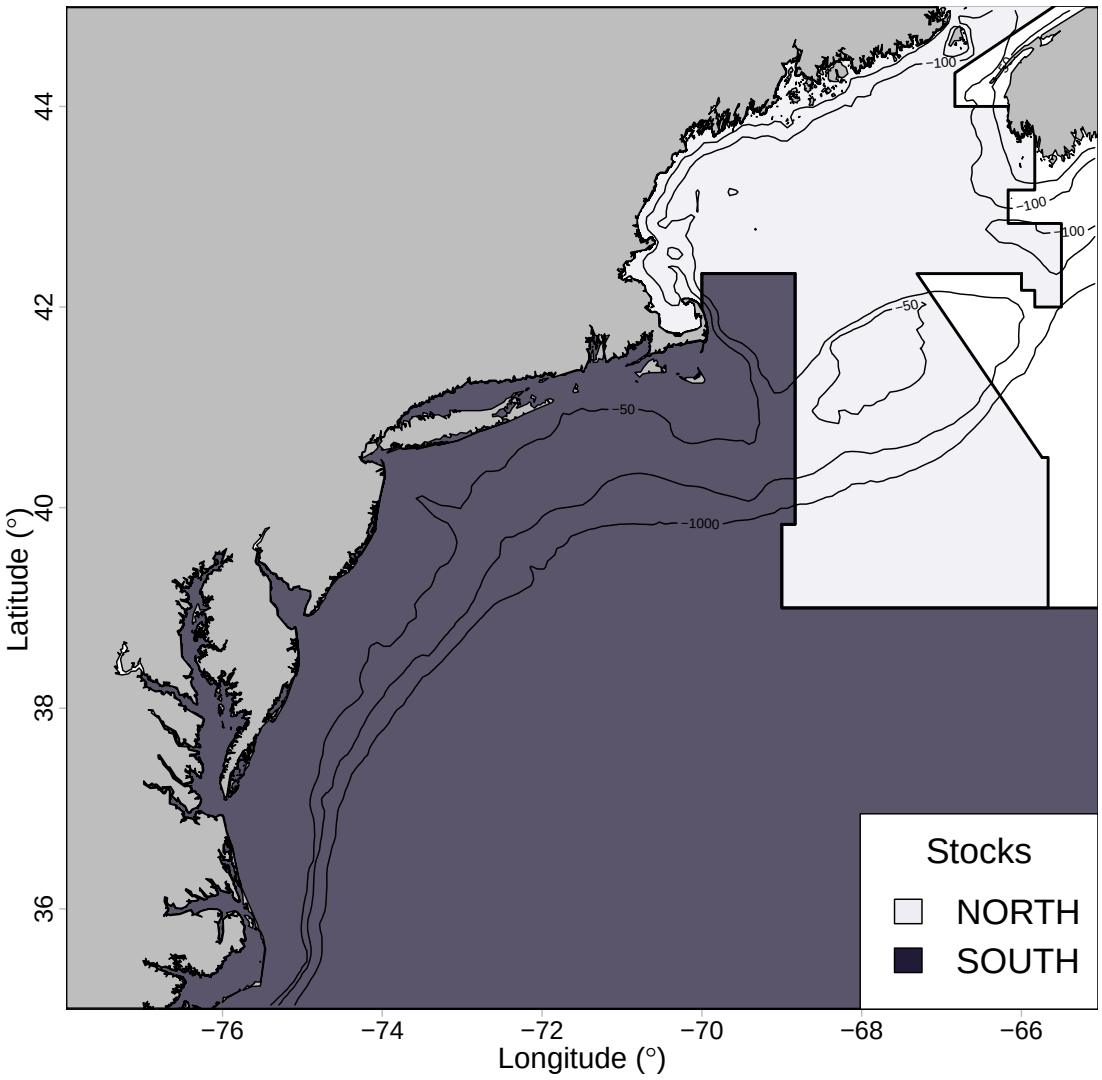


Figure 2: Ocean quahog Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

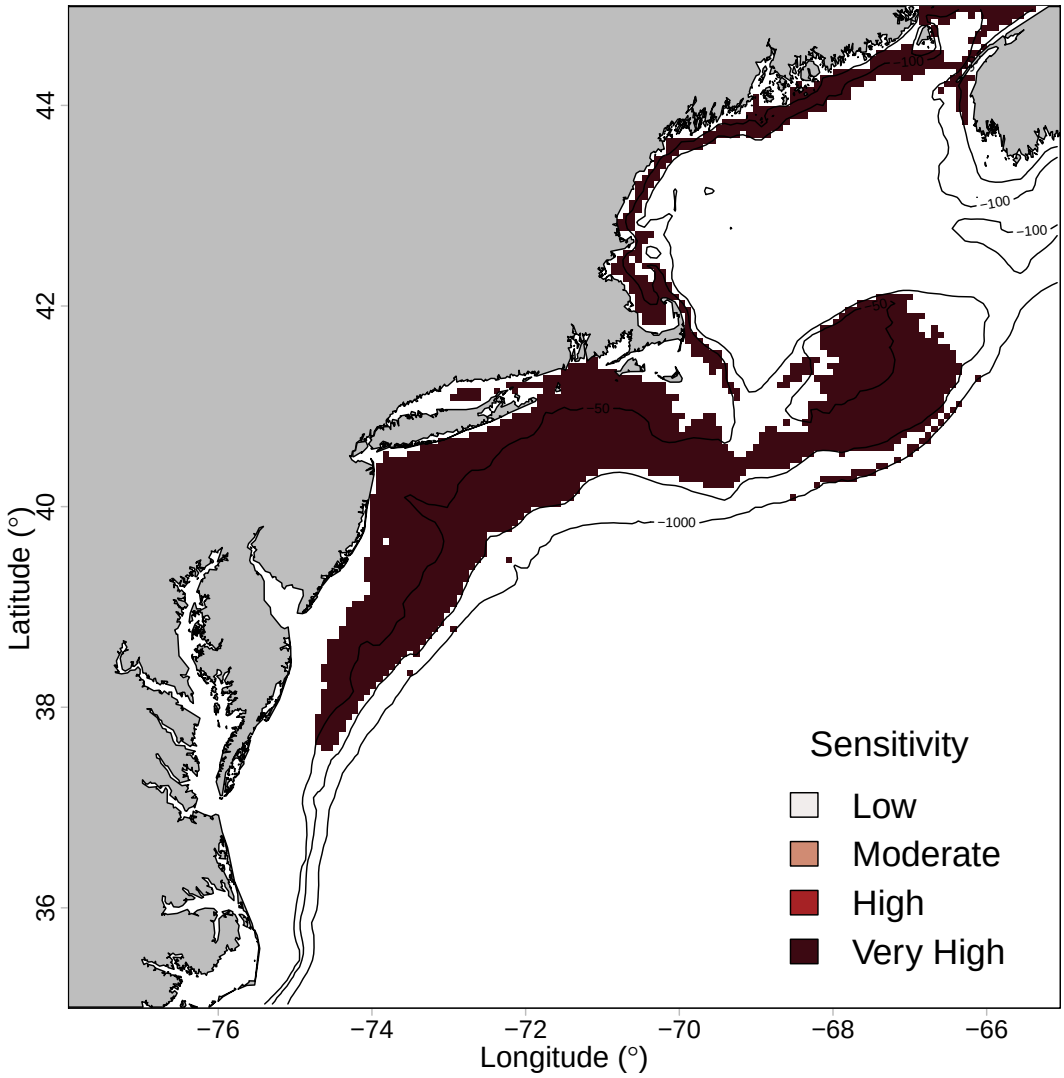


Figure 3: Ocean quahog Sensitivity on the Northeast Shelf

The ocean quahog received a **Very High** sensitivity score, with a certainty of 71%. Three attributes had scores greater than 3.5: Adult Mobility (4), Population Growth Rate (4), and Relationship to Calcified Organisms (3.6) (Figure 1).

Similar to other bivalves, ocean quahogs have calcium carbonate shells and are sessile (Cargnelli et al. 1999). They are also one of the longest-lived bivalve species, with maximum ages high as 507 years old documented, and ages over 300 years old common in the Northwest Atlantic (Sower et al. 2023). They mature at 13 years of age (Cargnelli et al. 1999).

Ocean quahogs also received a Very High sensitivity score in the previous Climate Vulnerability Assessment for the Northeast Shelf. The same attributes: Adult Mobility, Population Growth Rate, and Sensitivity to Ocean Acidification, drove the score (Hare et al. 2016).

Exposure:

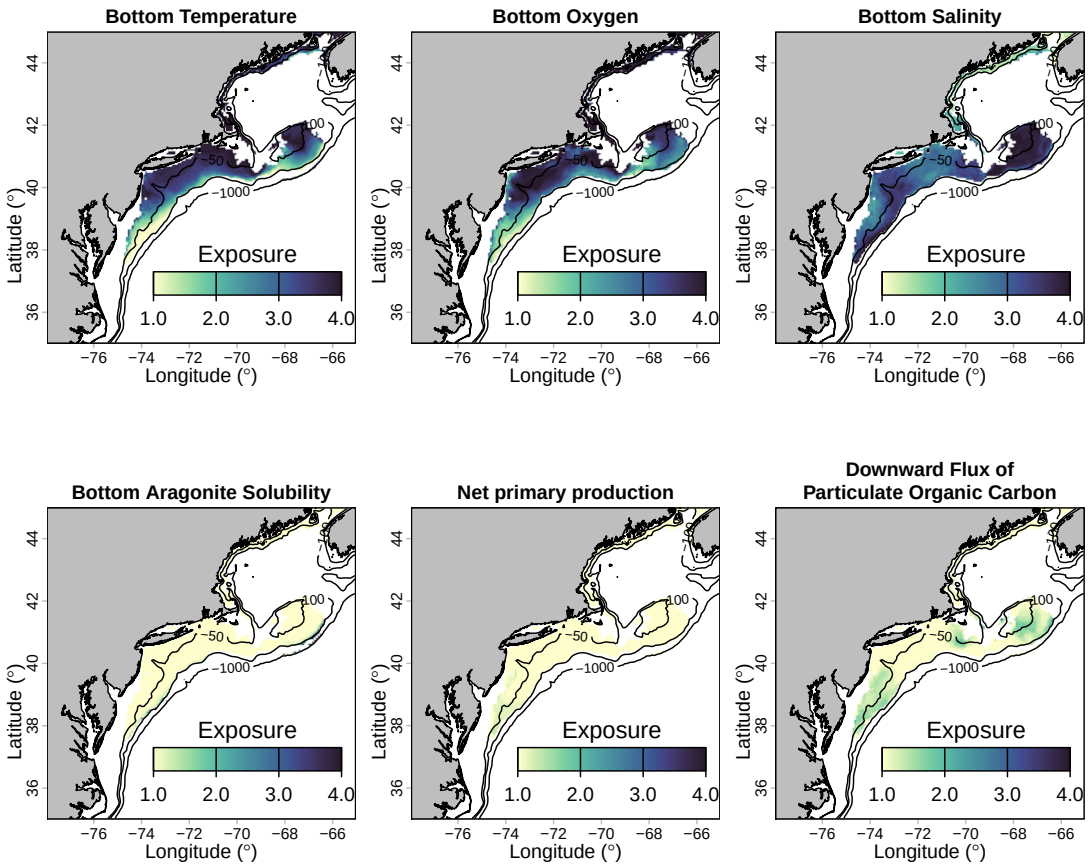


Figure 4: Maps of each Exposure Factor for Ocean quahog. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure scores for the ocean quahog were **Moderate** (2.46 +/- 0.90) (Figure 1). Exposure was Very High on Georges Bank, and High along the coast of the Gulf of Maine and in Southern New England. Scores were Low elsewhere in the distribution (Figure 5). These patterns were driven by bottom temperature, oxygen, and salinity exposure (Figure 4).

In the previous Assessment, ocean quahog received a High Exposure score, driven by high sea surface temperature and ocean acidification scores (Hare et al. 2016).

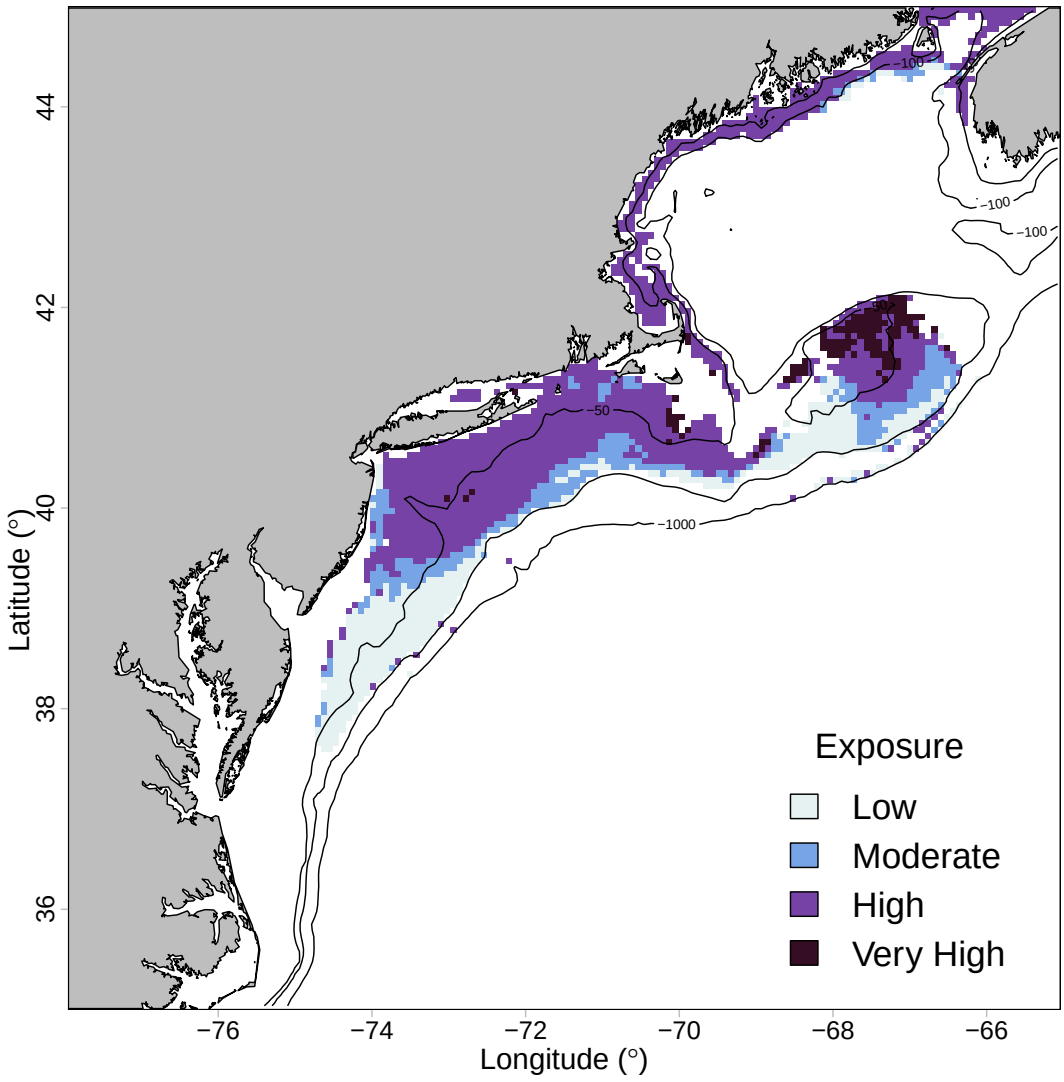


Figure 5: Total Exposure on the Northeast Shelf for Ocean quahog. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

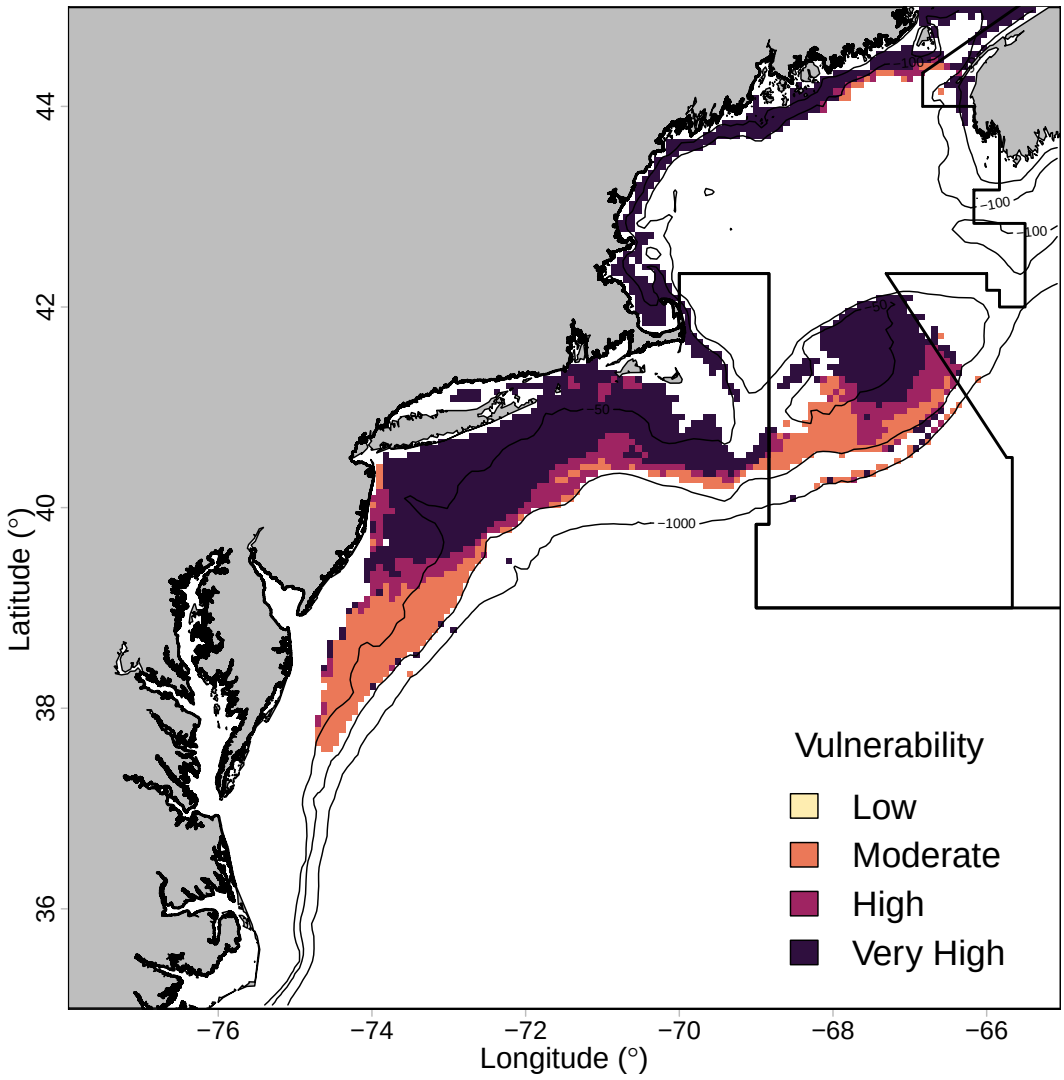


Figure 6: Vulnerability on the Northeast Shelf for Ocean quahog. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

Mean Vulnerability scores for ocean quahog were **High** (9.85 +/- 3.58) (Figure 1). Scores were Very High in Southern New England and on Georges Bank and Moderate elsewhere along the continental shelf (Figure 6).

In the previous Assessment, ocean quahog received a Very High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

87% of data quality scores were greater than or equal to 2, indicating strong data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The ensemble ocean quahog model received a 2.2 (+/- 0.2) mean model confidence score, indicating moderate confidence in the model. The predicted distributions matched experts' expectations, but the limited sample size decreased overall scores. The ensemble model AUC score was 0.88, indicating good model performance.

Additional Metrics

Distribution Shifts:

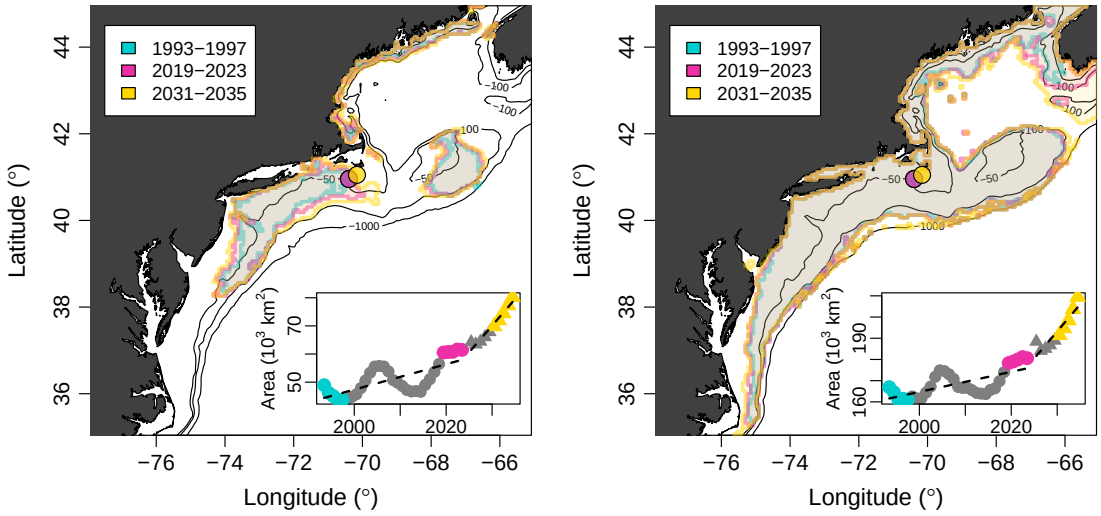


Figure 7: Distribution shifts on the Northeast Shelf for Ocean quahog. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Ocean quahog habitat area increased significantly between 1993 and 2023. Core habitat grew in southern New England and the Mid Atlantic Bight, as well as over Georges Bank, which the species range expanded southward (Figure 7). Because these shifts were relatively uniform, there was very little shift in the weighted centroid; the centroid shifted less than a kilometer to the south-southeast between 1993 and 2023.

Under near future ocean conditions, ocean quahog habitat is predicted to continue expanding. Habitat will continue to expand in Southern New England and over Georges Bank, but habitat is also predicted to expand along the coast of the Gulf of Maine (Figure 7). As a result, the weighted centroid is predicted to shift 22 km to the east-northeast by 2035.

In the previous Assessment, ocean quahog received a High Distributional Vulnerability Rank (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	[INSERT]	[INSERT]	[INSERT]	[INSERT]
2019-2023	[INSERT]	[INSERT]	[INSERT]	[INSERT]	[INSERT]
2031-2035	[INSERT]	[INSERT]	[INSERT]	[INSERT]	[INSERT]

Directional Effect:

Experts predicted that Ocean Quahog would be **negatively** impacted by climate change. Certainty was 100%. Warming conditions are expected to shift the species' distribution north and further offshore, possibly leading to competition between surfclams and ocean quahogs (Weinberg et al. 2002) (Stromp et al. 2023) (Spencer et al. 2024). Ocean acidification will also negatively impact the species. The 2016 Climate Vulnerability Assessment also suggested that Ocean Quahogs would be negatively impacted by climate change (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Radiocarbon aging on ocean quahog shells suggest past (200 years) ocean quahog distributions inshore of their current range, indicating a range shift coinciding with the regression of the Cold Pool after the Little Ice Age (LeClaire et al. 2022). With settlement occurring at the end of the larval stage, it may be presumed that temperature impacts settlement rate. Weinberg et al. (2002) note that an increase in temperature would shift the species offshore and north. Future warming conditions are predicted to result in range shifts in Atlantic surfclams that would lead to potential competition for food and space between surfclams and ocean quahogs (Stromp et al. 2023) (Spencer et al. 2024).

Life History Synopsis

Ocean quahogs are bivalve mollusks that inhabit the Northeast U.S. Continental Shelf from Newfoundland to Cape Hatteras (Cargnelli et al. 1999). The ocean quahog is among the longest-lived and slowest growing of marine bivalves, reaching maturity around 13 years of age and harvestable size at age 20 (Cargnelli et al. 1999). Maximum ages have been documented as high as 507 years old, with ages over 300 years old more common in the Northwest Atlantic (Sower et al. 2023). Spawning occurs either in summer or fall, and can extend for months, releasing eggs and sperm into the water column at low levels over time (Cargnelli et al. 1999). Multiple annual spawnings may occur at the individual and population levels (Mann 1982). Eggs and larvae drift with currents until metamorphosis, then settle to the bottom (Mid-Atlantic Fishery Management Council (MAFMC) 1997). Larval development, at minimum, lasts for 55 days and decreases with increasing temperature (Lutz et al. 1981). Larvae go through three stages of development (trochophores, veligers and pediveligers) with the duration of each stage being temperature dependent (Cargnelli et al. 1999). Van de Water Lewis (1997) noted that wind-forced circulation may play a significant role in planktonic and benthic recruitment on Georges Bank and that interannual differences in these winds can yield very different abundance patterns over the bank. Once settled to the bottom, juvenile ocean quahogs are found offshore in sandy substrates, but may survive in muddy intertidal environments if protected from predators (Kraus et al. 1989). Cross-shelf distributions of juveniles and

adults remain within a 10°C optimal temperature range, as temperatures above 16°C limit distribution (Cargnelli et al. 1999). Ocean quahogs are suspension feeders on phytoplankton, using their relatively short siphons which are extended above the surface of the substrate to pump in water (Cargnelli et al. 1999). Juvenile and small ocean quahogs are commonly preyed upon by crustaceans (rock crabs, sea stars) and fish (haddock, cod) (Cargnelli et al. 1999). Ocean quahogs are managed by the Mid-Atlantic Fishery Management Council under the Atlantic Surfclam-Ocean Quahog Fishery Management Plan. The species is mainly harvested off of Maine and Long Island. The last stock assessment was conducted in 2019, where the stock was not overfished and overfishing was not occurring (Mid-Atlantic Fishery Management Council (MAFMC) 2025).

Supplemental Scores

Exposure Timeseries

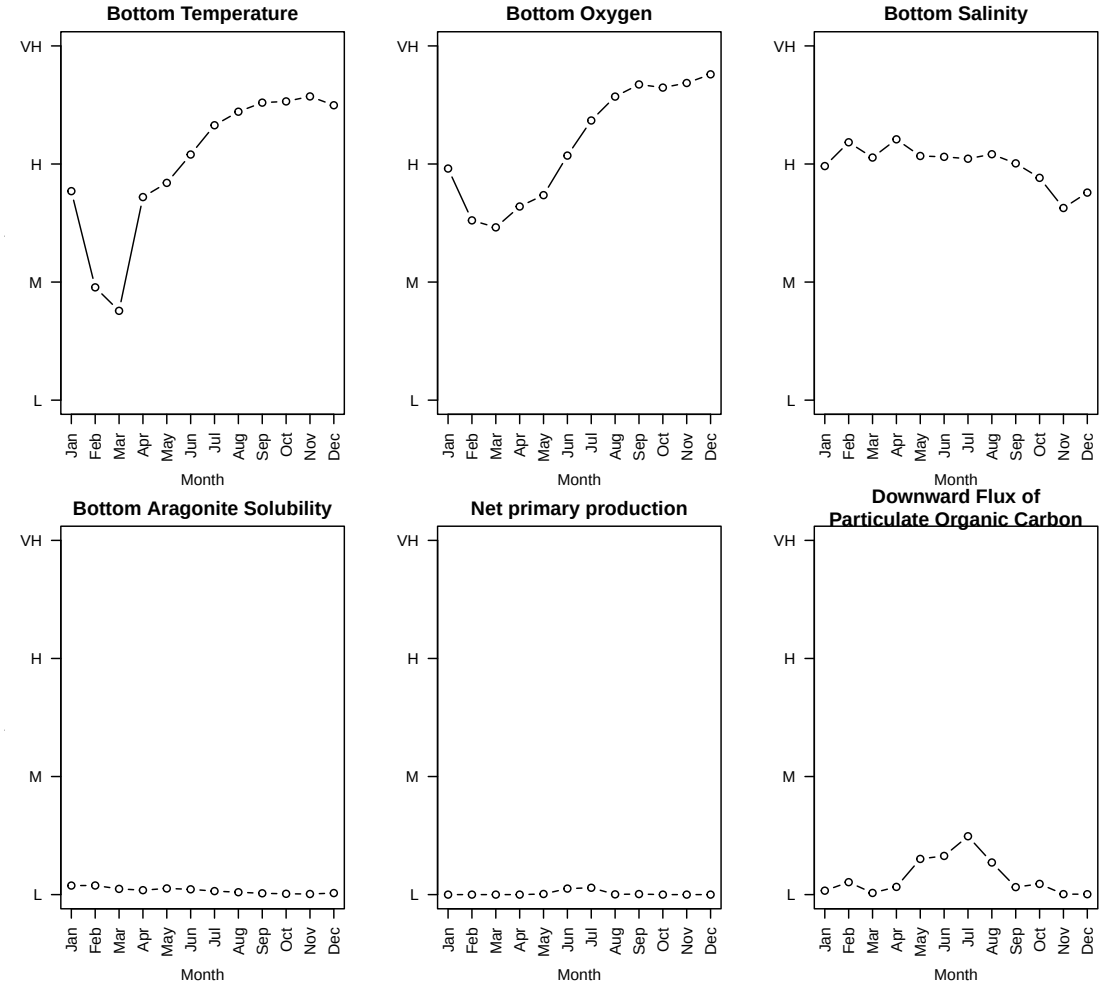


Figure 8: Monthly Timeseries of each Exposure Factor for Ocean quahog. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure scores ranged from Low to High. Scores were Low in March and High from June through December (Figure 9). These patterns were driven by changes in bottom temperature and oxygen exposure (Figure 8).

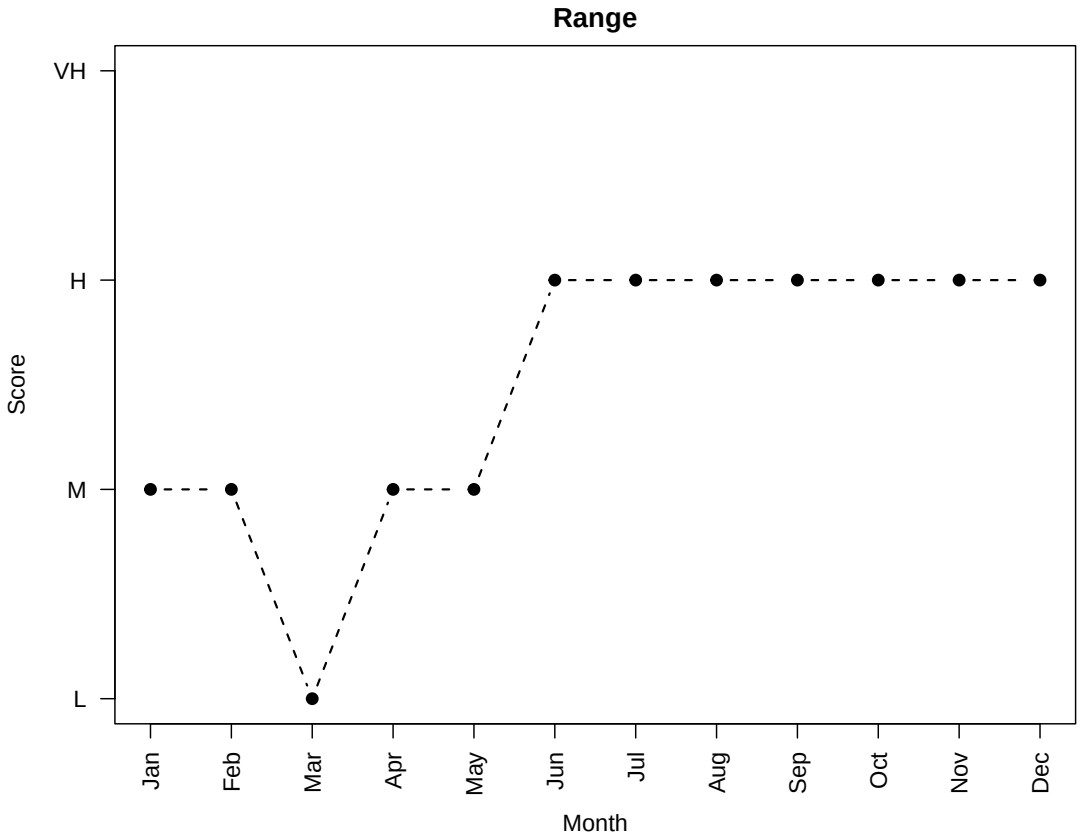


Figure 9: Timeseries of Total Exposure for Ocean quahog

Exposure due to Significant Factors

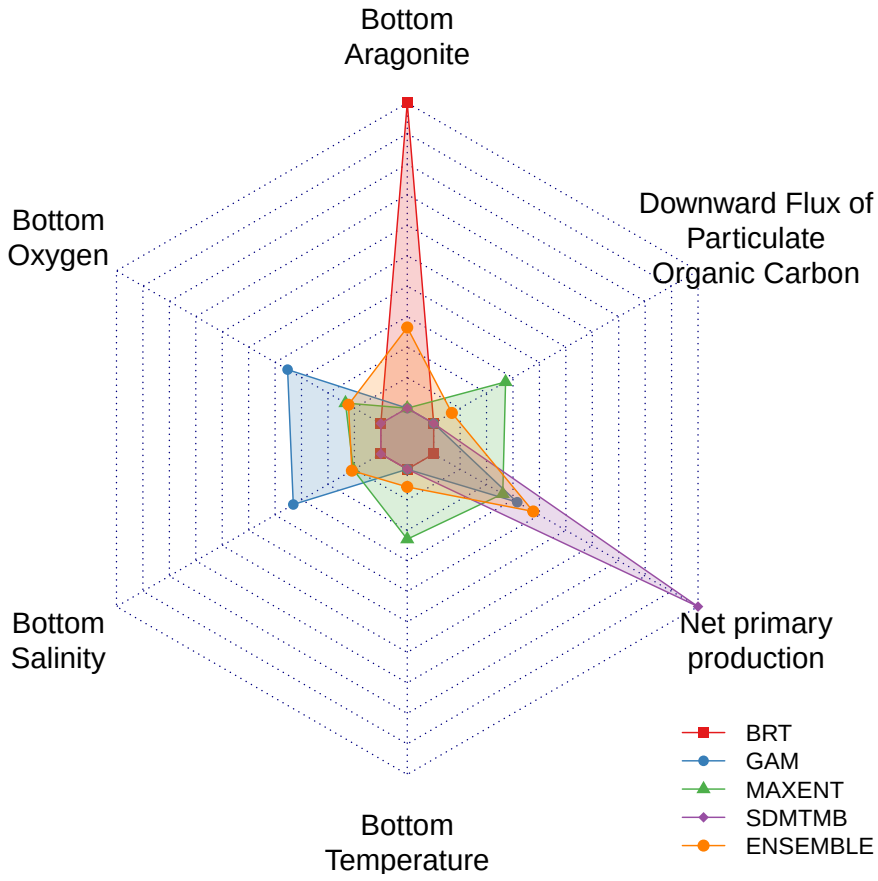


Figure 10: Dynamic Factors driving Ocean quahog distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom oxygen, salinity, aragonite solubility, and net primary productivity were the major dynamic drivers of ocean quahog distributions (Figure 10). As a benthic shellfish, they occupy the benthos and are filter feeders with calcium carbonate shells (Cargnelli et al. 1999). When only these factors were used to calculate exposure, exposure remained Moderate (1.86 +/- 0.83) (Figure 12). High exposure scores were found south of Long Island, Cape Cod, and on Georges Bank (Figure 11).

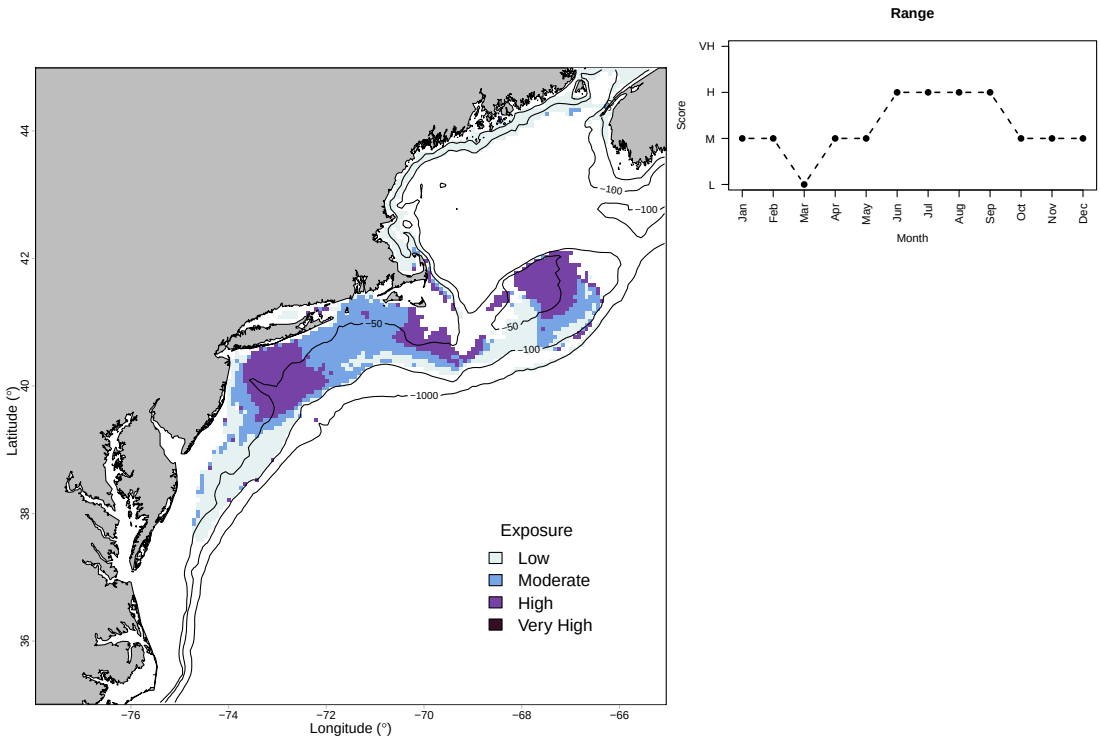
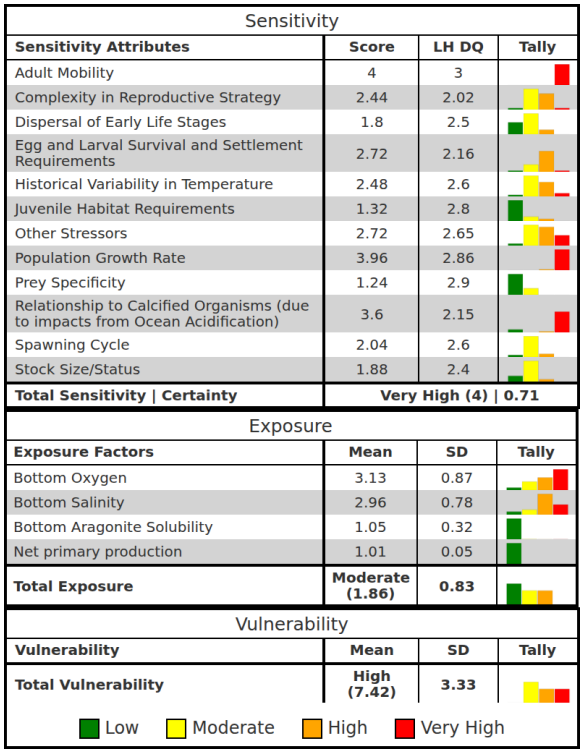


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Ocean quahog distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	4	3	
Complexity in Reproductive Strategy	2.44	2.02	
Dispersal of Early Life Stages	1.8	2.5	
Egg and Larval Survival and Settlement Requirements	2.72	2.16	
Historical Variability in Temperature	2.48	2.6	
Juvenile Habitat Requirements	1.32	2.8	
Other Stressors	2.72	2.65	
Population Growth Rate	3.96	2.86	
Prey Specificity	1.24	2.9	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.6	2.15	
Spawning Cycle	2.04	2.6	
Stock Size/Status	1.88	2.4	
Total Sensitivity | Certainty	**Very High (4) | 0.71**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Oxygen	3.13	0.87	
Bottom Salinity	2.96	0.78	
Bottom Aragonite Solubility	1.05	0.32	
Net primary production	1.01	0.05	
Total Exposure	**Moderate (1.86)**	**0.83**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**High (7.42)**	**3.33**	

Low Moderate High Very High

Figure 12: As in Figure 1 for Ocean quahog, but with only significant Exposure factors and the resulting Vulnerability Scores.

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